

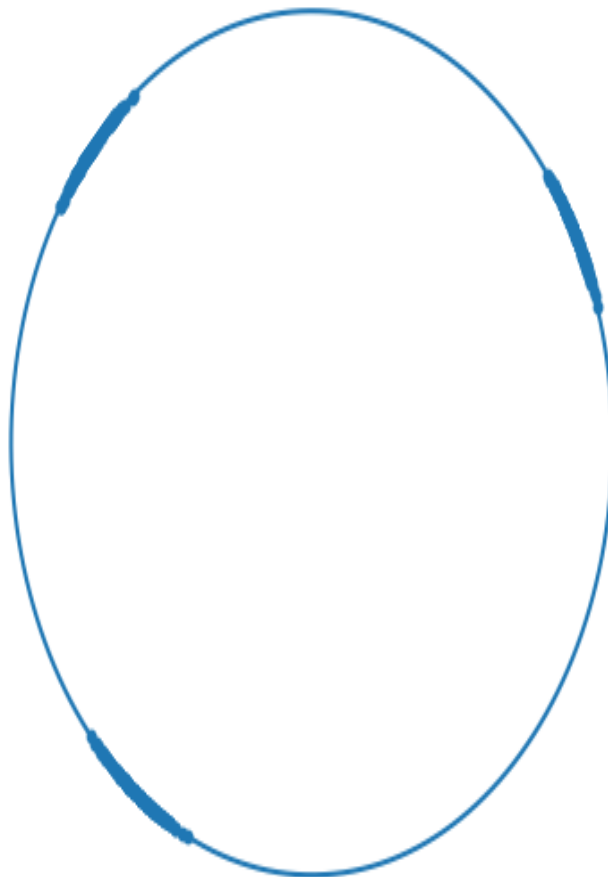
Angular Manifold Routing — Research Explainer (Version 4)

This document presents a geometric routing mechanism designed to improve the computational efficiency of large AI systems. The core discovery is that when embeddings are normalized and routed purely by direction on a hypersphere, tokens naturally cluster into far fewer routing buckets than expected. This produces a scalable form of sparse computation.

1. Directional Clustering of Embeddings

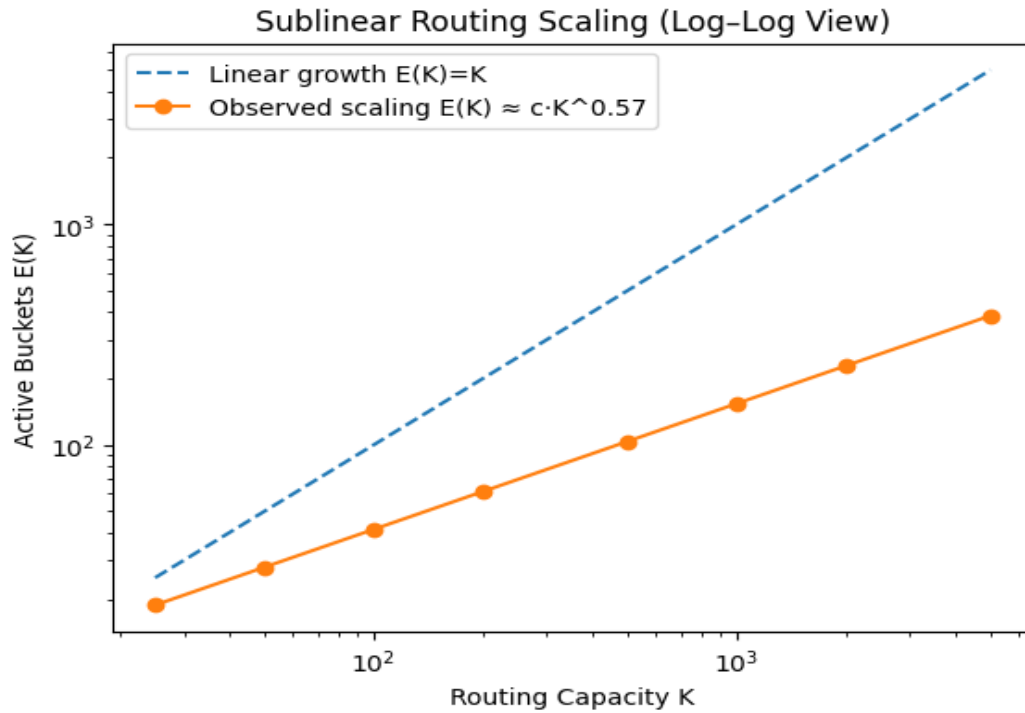
Embeddings representing related concepts tend to occupy nearby directions in vector space. After normalization, embeddings lie on the unit hypersphere S^{d-1} . Routing by direction causes tokens with similar semantics to cluster into shared compute sectors.

Embedding Directions Cluster on the Unit Hypersphere (2D Analogy)



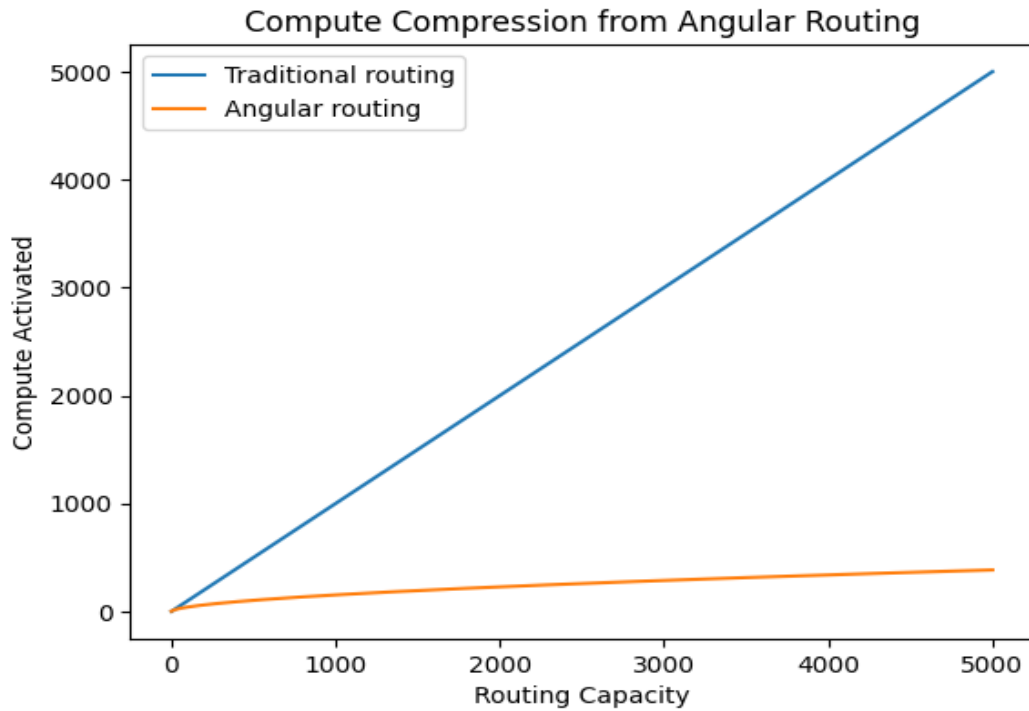
2. Empirical Scaling Law

Experiments indicate that the number of active routing buckets grows sublinearly with routing capacity. The empirical relationship can be written as: $E(K) \approx c \cdot K^{0.57}$ where K is routing capacity and $E(K)$ is the number of active compute groups.



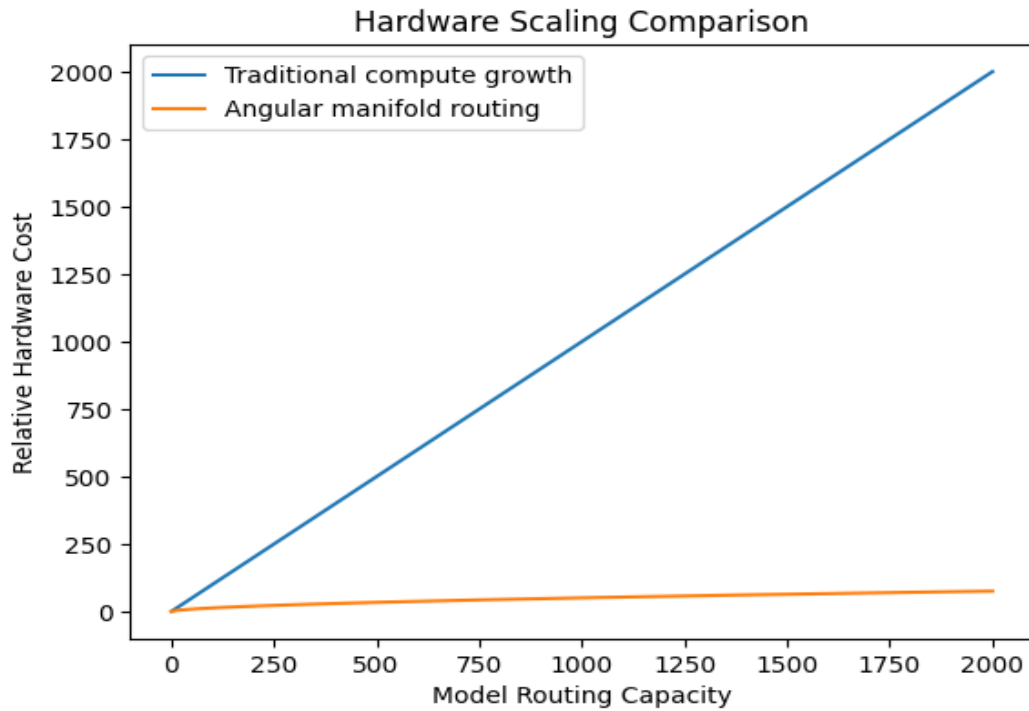
3. Compression Effect

Because $E(K)$ grows slower than K , increasing router capacity does not increase compute cost linearly. The router effectively compresses routing capacity into a smaller set of active compute groups.



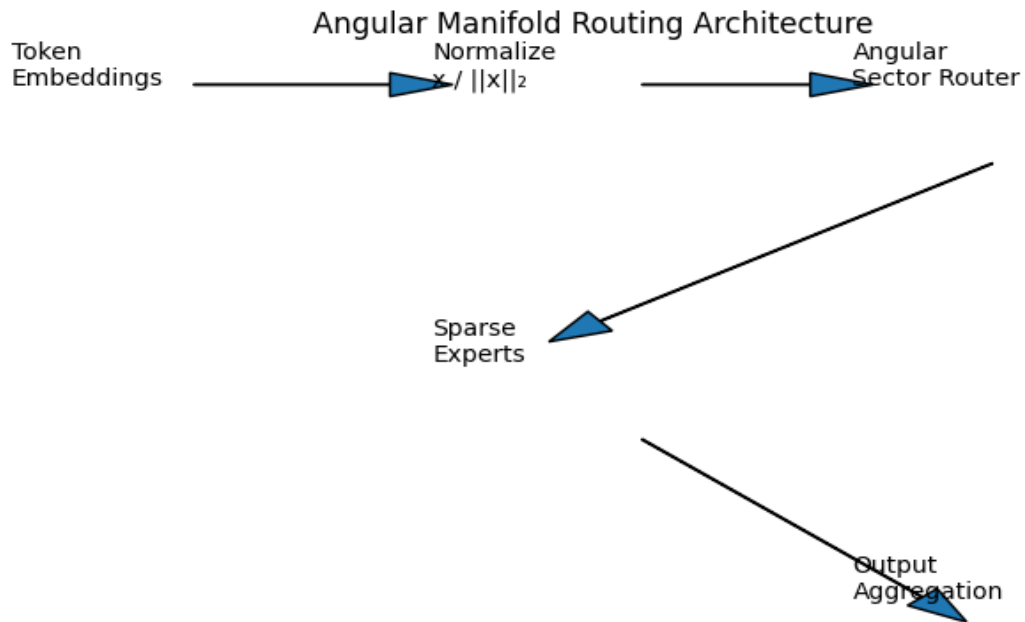
4. Hardware Implications

Sublinear routing growth leads to improved hardware efficiency. Larger routing capacity can be supported while activating fewer compute pathways, reducing memory traffic and compute cost.



5. Architecture Overview

The routing pipeline is conceptually simple: 1. Token embeddings are produced. 2. Embeddings are normalized to the unit hypersphere. 3. The hypersphere is partitioned into angular sectors. 4. Tokens are routed according to direction. 5. Only occupied sectors activate compute experts. 6. Expert outputs are aggregated. The geometry of the embedding space organizes routing automatically, removing the need for complex learned gating.



Summary

Angular Manifold Routing suggests that geometric structure in embedding space can produce natural computational sparsity. Directional clustering leads to sublinear routing growth, which could enable more scalable and efficient AI systems.